

CS-AML

Software Requirements Specification (SRS)

Version 0.1

Document status

Normative software baseline for MVP 0.1 implementation. This SRS translates the approved PRD, Product & Feature Specification, Data Model Specification, Control Implementation Guide, Investigation Methodology, and Technology Architecture into testable software requirements.

Civil Society Financial Intelligence / AML Investigation Platform

Document Control

Document	CS-AML Software Requirements Specification
Version	0.1
Status	Normative baseline
Primary product scope	MVP 0.1
Audience	Product, engineering, QA, security, data, reviewers, governance
Upstream documents	CS-AML PRD v0.1; Product & Feature Specification v0.1; Data Model Specification v0.1; Technology Architecture v0.1; Control Implementation Guide v0.1; Investigation Methodology v0.1
Normative terms	SHALL / MUST = mandatory; SHOULD = recommended; MAY = optional

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1. Purpose and Scope

This SRS specifies the software behaviour, data obligations, interfaces, controls, quality attributes, and verification conditions required to implement CS-AML MVP 0.1 as a secure system of record for civil-society financial-intelligence investigations.

Primary acceptance objective

A small two-analyst team SHALL be able to open a sensitive case, ingest and preserve evidence, resolve entities, map relationships/assets/events/value flows, test competing hypotheses, assess typology consistency and confidence, complete independent review, and export an approved intelligence product while retaining provenance and audit history.

1.1 In scope

- Case and workflow management
- Source/evidence/provenance management
- Document ingestion
- Entity registry and reversible entity resolution
- Relationships, ownership, control, and assets
- Events and timeline
- Value-flow modelling
- Typology and indicator analysis
- Hypothesis, gap, confidence, and assessment management
- Search and graph exploration
- Intelligence products, review, dissemination and sharing logs
- IAM, protected-source compartment, audit, retention, backup and restore

1.2 Out of scope for MVP 0.1

- Automated AML transaction monitoring against banking core systems
- Autonomous guilt determination or law-enforcement decisioning
- Continuous web-scale scraping
- Cross-organisation federation
- Advanced blockchain tracing
- Automated sanctions/PEP screening as a mandatory workflow
- Advanced machine-learning risk scoring
- Multi-tenant SaaS operation

2. Product and System Context

CS-AML is positioned as a civil-society financial-intelligence and investigation platform, not a bank compliance engine. The software converts structured investigative work into reproducible intelligence while preserving uncertainty and provenance.

2.1 Product axiom

Axiom

Technology SHALL preserve analytical uncertainty rather than erase it. Source, evidence, claim, fact, indicator, hypothesis, assessment, and intelligence product remain distinguishable in storage, APIs, UI, graph views, automation, and exports.

2.2 Upstream analytical chain

SOURCE → EVIDENCE → CLAIM/FACT → INDICATOR → HYPOTHESIS → ASSESSMENT → INTELLIGENCE PRODUCT

2.3 Mandatory value-flow classes

DIRECT | DOCUMENTED | RECONSTRUCTED | HYPOTHETICAL

3. Definitions and Conventions

Term	Meaning	Software implication
Canonical object	Authoritative record of investigative state	Must be durable, versioned where material, access-controlled, and auditable
Derived projection	Rebuildable search/graph/cache/AI representation	Must never become sole source of truth
Material action	Action capable of changing analytical meaning, access, review status, or dissemination	Must be auditable
Protected source	Human/source identity requiring compartmentalisation	Identity access separate from routine evidence access
High-impact product	Product containing potentially harmful adverse findings or public attribution	Requires independent review and dissemination approval
MVP	Minimum release satisfying end-to-end case completion	All P0 requirements required unless explicitly waived

4. System Boundary and Actors

4.1 Primary actors

Actor	Responsibility
Investigator	Creates and develops case material
Case Owner	Accountable for scope, gates, and dissemination decisions
Analyst	Performs entity/value-flow/typology/hypothesis analysis
Reviewer	Independent quality review
Evidence Custodian	Preserves originals, integrity, and evidence lineage
Data Steward	Approves significant entity merge/unmerge and data-quality changes
Source Handler	Controls protected-source identity
Platform/Security Admin	Operates IAM, configuration, backup, monitoring
Auditor	Reviews controls and audit history

4.2 Trust boundaries

- User browser/client
- Application/API policy layer
- Canonical database
- Evidence object store
- Protected-source compartment
- Search/graph projections
- External integrations
- Backup/restore environment
- Administrative control plane

5. Architectural Constraints

- Canonical data and evidence remain independent from rebuildable search/graph projections.
- Original evidence is immutable to routine analyst workflows; transformations create derivatives.
- Entity merge decisions are reversible and must preserve history.
- All material adverse analytical claims remain traceable to evidence.
- Authorization is enforced at API/service layer, not only UI.
- High-impact dissemination requires accountable human authorization.
- AI and automation may assist but cannot silently promote generated output into material facts.

6. Functional Requirements

6.1 Case and Workflow

SRS-FR-CASE-001 — Case creation

Requirement	The system SHALL create a stable Case ID and require title, accountable owner, purpose, investigation question, sensitivity/classification, and status before a case becomes Active.
Rationale	Prevents orphaned or undefined investigations.
Verification	Create incomplete case and verify activation is blocked; complete required fields and verify activation succeeds.
Priority	MVP / P0
Traceability	F-CASE-001; GOV-01; CAS-01

SRS-FR-CASE-002 — Investigation Charter

Requirement	The system SHALL version the Investigation Charter, including question, scope, exclusions, jurisdictions, period, authorised/prohibited collection methods, risks, and intended outputs.
Rationale	Controls scope and mission creep.
Verification	Edit charter twice and verify prior version remains retrievable with actor/time.
Priority	MVP / P0
Traceability	F-CASE-002; CAS-01; PRI-01

SRS-FR-CASE-003 — Lifecycle gates

Requirement	The system SHALL support configurable G0–G6 lifecycle gates and SHALL prevent controlled actions when required approval is absent.
Rationale	Enforces stage-gated investigation methodology.
Verification	Attempt high-impact transition without approval and verify denial; approve independently and retry.
Priority	MVP / P0
Traceability	F-CASE-003; CAS-02; QUA-01

SRS-FR-CASE-004 — Tasks and assignments

Requirement	The system SHALL support case-scoped tasks with owner, due date, type, status, dependencies, and immutable completion history.
Rationale	Coordinates investigative work.

Verification	Create, reassign, complete task and verify audit trail.
Priority	MVP / P0
Traceability	F-CASE-004

SRS-FR-CASE-005 — Case activity timeline

Requirement	The system SHALL display material case actions in chronological order with actor, timestamp, object, and action type.
Rationale	Supports reconstructability.
Verification	Perform material actions and verify ordered visibility.
Priority	MVP / P0
Traceability	F-CASE-005; AUD-01

6.2 Source, Evidence and Documents

SRS-FR-EVD-001 — Source register

Requirement	The system SHALL register material sources with origin, type, publisher/owner where known, access method, access timestamp, URL/location, reliability, legal/access note, and archival reference where available.
Rationale	Maintains provenance.
Verification	Create source and verify mandatory provenance validation.
Priority	MVP / P0
Traceability	F-EVD-001; SRC-01

SRS-FR-EVD-002 — Original preservation

Requirement	The system SHALL preserve original evidence separately from working/derived artefacts and SHALL prevent routine analysts from modifying the original object.
Rationale	Protects evidence integrity.
Verification	Upload file, attempt overwrite as analyst, verify denial and new-version workflow.
Priority	MVP / P0
Traceability	F-EVD-002; EVD-01

SRS-FR-EVD-003 — Cryptographic integrity

Requirement	The system SHALL compute SHA-256 or stronger approved hash for critical uploaded evidence and SHALL support recomputation/verification.
Rationale	Detects silent change.

Verification	Recompute hash and compare; alter test copy and verify mismatch.
Priority	MVP / P0
Traceability	F-EVD-003; EVD-01

SRS-FR-EVD-004 — Evidence extracts

Requirement	The system SHALL allow page/paragraph/region extracts that remain linked to parent evidence and a precise source location.
Rationale	Enables citations and review.
Verification	Create extract and navigate back to original location.
Priority	MVP / P0
Traceability	F-EVD-004; EVD-02

SRS-FR-EVD-005 — Derivative lineage

Requirement	The system SHALL record parent evidence, transformation type, tool/version where material, creator, and timestamp for OCR, translation, crop, parse, or derived dataset.
Rationale	Preserves transformation lineage.
Verification	Create derivative and verify reverse trace to original.
Priority	MVP / P0
Traceability	F-EVD-005; EVD-02

SRS-FR-EVD-006 — Reliability and credibility

Requirement	The system SHALL store source reliability independently from information credibility and SHALL not reuse one value for both dimensions.
Rationale	Prevents false certainty.
Verification	Save different ratings and verify independent display/storage.
Priority	MVP / P0
Traceability	F-EVD-006; SRC-02; ASM-01

SRS-FR-DOC-001 — Document ingestion

Requirement	The system SHALL ingest PDF, DOCX, XLSX, CSV, TXT, and common image formats while preserving original bytes.
Rationale	Supports core evidence intake.
Verification	Upload representative formats and verify originals plus metadata.
Priority	MVP / P0

Traceability	F-DOC-001
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6.3 Entity, Relationship and Asset

SRS-FR-ENT-001 — Entity registry

Requirement	The system SHALL support canonical Entity records for person, organisation/company, account, wallet, address, domain, phone/email, government agency, and extensible types.
Rationale	Provides reusable identity layer.
Verification	Create each baseline type and verify unique stable identifiers.
Priority	MVP / P0
Traceability	F-ENT-001; ENT-01

SRS-FR-ENT-002 — Aliases and identifiers

Requirement	The system SHALL allow conflicting aliases/identifiers to coexist with separate provenance and status.
Rationale	Preserves uncertainty and contradictory data.
Verification	Add conflicting identifiers and verify neither overwrites the other.
Priority	MVP / P0
Traceability	F-ENT-002

SRS-FR-ENT-003 — Candidate matching

Requirement	The system SHALL present matching and conflicting features for possible duplicates before merge.
Rationale	Reduces false entity consolidation.
Verification	Create similar entities and verify comparison view.
Priority	MVP / P0
Traceability	F-ENT-003; ENT-01

SRS-FR-ENT-004 — Merge and unmerge

Requirement	The system SHALL support evidence-based merge with rationale and SHALL support unmerge that restores prior records and relationships without erasing history.
Rationale	Protects analytical reversibility.
Verification	Merge then unmerge test entities; compare pre/post state and audit log.
Priority	MVP / P0

Traceability	F-ENT-004; AUD-01
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SRS-FR-ENT-005 — Entity resolution status

Requirement	The system SHALL support candidate, probable, confirmed, disputed, and unresolved statuses and SHALL expose status in search/graph views.
Rationale	Keeps identity uncertainty visible.
Verification	Change status and verify graph/search labels.
Priority	MVP / P0
Traceability	F-ENT-005

SRS-FR-REL-001 — First-class relationship

Requirement	The system SHALL represent material relationships as canonical objects with type, endpoints, evidence, dates, confidence, status, and analyst.
Rationale	Prevents graph-only unsupported edges.
Verification	Create relationship and trace from graph to evidence.
Priority	MVP / P0
Traceability	F-REL-001; REL-01

SRS-FR-REL-002 — Ownership interest

Requirement	The system SHALL represent direct/indirect ownership with percentage or unknown, dates, and evidence.
Rationale	Supports beneficial-ownership analysis without fabricated zeroes.
Verification	Create unknown and numeric ownership records; verify semantics.
Priority	MVP / P0
Traceability	F-REL-002

SRS-FR-REL-003 — Control distinction

Requirement	The system SHALL distinguish legal ownership, beneficial ownership, control, use, and association.
Rationale	Avoids collapsing materially different relationships.
Verification	Create each relation type and verify distinct UI/API representation.
Priority	MVP / P0
Traceability	F-REL-003; AST-01

SRS-FR-AST-001 — Asset registry

Requirement	The system SHALL support property, vehicle, shares, vessel, aircraft, crypto and extensible asset types with attribution type and evidence.
Rationale	Supports asset tracing.
Verification	Create asset with ownership and control assertions.
Priority	MVP / P0
Traceability	F-AST-001

6.4 Event, Timeline and Value Flow

SRS-FR-TIM-001 — Event records

Requirement	The system SHALL store events with type, date/date-range, date precision, linked entities/assets, location where relevant, and evidence.
Rationale	Supports temporal analysis.
Verification	Create exact/approximate events and verify precision retained.
Priority	MVP / P0
Traceability	F-TIM-001

SRS-FR-TIM-002 — Visual timeline

Requirement	The system SHALL render filtered timelines for case/entity/asset and link each item to canonical event/evidence.
Rationale	Enables chronological reconstruction.
Verification	Filter timeline and open source event.
Priority	MVP / P0
Traceability	F-TIM-002

SRS-FR-VAL-001 — Value-flow record

Requirement	The system SHALL require origin, destination, mechanism, value/range/unknown, currency where applicable, date/date-range, evidence, confidence, and mandatory flow_class.
Rationale	Makes value movement explicit and qualified.
Verification	Create all four flow classes and verify mandatory fields.
Priority	MVP / P0
Traceability	F-VAL-001; VAL-01

SRS-FR-VAL-002 — Multi-leg flows

Requirement	The system SHALL support multi-leg value-flow chains in which every leg retains separate evidence, class, value, and confidence.
Rationale	Avoids certainty inheritance across a chain.
Verification	Create 3-leg mixed-class flow and inspect each leg.
Priority	MVP / P0
Traceability	F-VAL-002

SRS-FR-VAL-003 — Flow visualisation

Requirement	The system SHALL visually distinguish DIRECT, DOCUMENTED, RECONSTRUCTED, and HYPOTHETICAL flows in interactive and exported views.
Rationale	Prevents inference from looking like direct transaction evidence.
Verification	Export mixed-class flow and verify class distinction persists.
Priority	MVP / P0
Traceability	F-VAL-003

SRS-FR-VAL-004 — Unknown and ranges

Requirement	The system SHALL represent unknown, minimum, maximum, and approximate values without coercing unknown to numeric zero.
Rationale	Preserves uncertainty.
Verification	Store unknown/range values and inspect API/UI.
Priority	MVP / P0
Traceability	F-VAL-004

6.5 Typology, Hypothesis and Assessment

SRS-FR-TYP-001 — Typology catalogue

Requirement	The system SHALL expose versioned CS-AML typology entries with definition, mechanism, observables, indicators, counter-indicators, false positives, and references.
Rationale	Supports consistent analysis.
Verification	Open typology and verify version and fields.
Priority	MVP / P0
Traceability	F-TYP-001; TYP-01

SRS-FR-TYP-002 — Indicator capture

Requirement	The system SHALL allow indicators and counter-indicators to link to evidence and analytical objects.
Rationale	Prevents free-floating allegations.
Verification	Create indicator with evidence and verify assessment traceability.
Priority	MVP / P0
Traceability	F-TYP-002

SRS-FR-TYP-003 — Typology match worksheet

Requirement	The system SHALL support controlled consistency levels and SHALL prevent a single weak indicator from automatically creating a strong typology match.
Rationale	Controls overstatement.
Verification	Create single weak indicator and verify no automatic strong match.
Priority	MVP / P0
Traceability	F-TYP-003

SRS-FR-HYP-001 — Competing hypotheses

Requirement	The system SHALL support at least two competing hypotheses per material investigation where alternatives are plausible.
Rationale	Reduces confirmation bias.
Verification	Create H1/H2 and link supporting/contradicting evidence.
Priority	MVP / P0
Traceability	F-HYP-001; HYP-01

SRS-FR-HYP-002 — Support and contradiction matrix

Requirement	The system SHALL store evidence/indicator effect on each hypothesis as support, contradiction, neutral, or unknown with analyst rationale.
Rationale	Makes reasoning inspectable.
Verification	Populate matrix and verify every cell retains rationale/history.
Priority	MVP / P0
Traceability	F-HYP-002

SRS-FR-HYP-003 — Intelligence gaps

Requirement	The system SHALL maintain explicit intelligence gaps
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	capable of affecting an assessment.
Rationale	Keeps unknowns visible.
Verification	Create gap and include it in final assessment.
Priority	MVP / P0
Traceability	F-HYP-003; GAP-01

SRS-FR-ASM-001 — Assessment

Requirement	The system SHALL store judgement, confidence, basis, alternatives, gaps, author, reviewer, and version.
Rationale	Creates reproducible analytical product.
Verification	Create and revise assessment; verify version history.
Priority	MVP / P0
Traceability	F-ASM-001; ASM-01

SRS-FR-ASM-002 — Confidence model

Requirement	The system SHALL support High, Moderate, Low, and Insufficient Basis (or configured equivalent) with rationale.
Rationale	Avoids pseudo-precision.
Verification	Save each level and verify mandatory rationale.
Priority	MVP / P0
Traceability	F-ASM-002

SRS-FR-ASM-003 — Backward traceability

Requirement	The system SHALL allow a reviewer to traverse an assessment backward through hypotheses/indicators/facts/evidence/sources.
Rationale	Core reproducibility requirement.
Verification	Select key assessment and trace to origin without external notes.
Priority	MVP / P0
Traceability	F-ASM-003

6.6 Search and Graph

SRS-FR-SCH-001 — Full-text and object search

Requirement	The system SHALL search authorised cases, entities, sources, evidence metadata, extracts, assets, events, and products using full-text and structured filters.
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Rationale	Supports investigation retrieval.
Verification	Search known fixture and verify expected scoped results.
Priority	MVP / P0
Traceability	F-SCH-001

SRS-FR-SCH-002 — Permission-filtered results

Requirement	The system SHALL apply object-level authorization before returning search results or counts.
Rationale	Prevents existence leakage.
Verification	Compare user with/without access and verify hidden objects do not affect result counts.
Priority	MVP / P0
Traceability	F-SCH-002; SEC-01

SRS-FR-GRF-001 — Graph exploration

Requirement	The system SHALL display authorised entities and canonical relationships with evidence/status/confidence access from each edge.
Rationale	Supports network analysis without hiding provenance.
Verification	Expand graph and open edge evidence.
Priority	MVP / P0
Traceability	F-GRF-001

6.7 Intelligence Product, Review and Dissemination

SRS-FR-PRD-001 — Product templates

Requirement	The system SHALL generate Financial Intelligence Note, Entity Profile, Asset Profile, Network Analysis, Referral Package, and Case Report from canonical objects.
Rationale	Standardises outputs.
Verification	Generate each baseline template with metadata/evidence index.
Priority	MVP / P0
Traceability	F-PRD-001

SRS-FR-REV-001 — Peer review workflow

Requirement	The system SHALL support comment, request-change, approve, and reject; high-impact approval SHALL be attributable and independent where policy requires.
Rationale	Implements quality gate.

Verification	Author attempts self-approval under independent-review policy and is denied.
Priority	MVP / P0
Traceability	F-REV-001; QUA-01

SRS-FR-PRD-002 — Versioning and corrections

Requirement	The system SHALL preserve prior intelligence-product versions and mark superseded/corrected versions explicitly.
Rationale	Prevents historical rewriting.
Verification	Issue correction and verify old version remains immutable/readable.
Priority	MVP / P0
Traceability	F-PRD-002; AUD-01

SRS-FR-PRD-003 — Evidence index

Requirement	The system SHALL generate an evidence/source index sufficient to trace every material fact/findings used in a product.
Rationale	Supports review and referral.
Verification	Generate product and click/resolve each material citation.
Priority	MVP / P0
Traceability	F-PRD-003

SRS-FR-DIS-001 — Dissemination approval

Requirement	The system SHALL block external export until required handling classification, recipient, purpose, and approval are recorded.
Rationale	Controls harm and confidentiality.
Verification	Attempt export before approval and verify denial.
Priority	MVP / P0
Traceability	F-DIS-001; DIS-01

SRS-FR-DIS-002 — Secure export

Requirement	The system SHALL export only explicitly approved objects and SHALL support minimisation/redaction.
Rationale	Prevents accidental over-disclosure.
Verification	Export package and verify excluded restricted object is absent.
Priority	MVP / P0

Traceability	F-DIS-002; PRI-01
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SRS-FR-DIS-003 — Referral package

Requirement	The system SHALL generate structured referral containing subjects, key facts, evidence references, timeline, value flows, indicators, gaps, confidence, and contact point.
Rationale	Makes outputs actionable.
Verification	Generate referral and verify fact/analysis/unknown sections remain distinct.
Priority	MVP / P0
Traceability	F-DIS-003

SRS-FR-DIS-004 — Sharing log

Requirement	The system SHALL record recipient, product/version, date, purpose, restrictions, and approving user for every external dissemination.
Rationale	Maintains accountability.
Verification	Share export and verify immutable log entry.
Priority	MVP / P0
Traceability	F-DIS-004; AUD-01

6.8 Administration, Security and Operations

SRS-FR-ADM-001 — Controlled vocabularies

Requirement	The system SHALL version relationship types, asset types, statuses, classifications, and other controlled vocabularies.
Rationale	Preserves historical semantics.
Verification	Change label and verify historical record retains version context.
Priority	MVP / P0
Traceability	F-ADM-001

SRS-FR-ADM-003 — Retention policies

Requirement	The system SHALL support retention/disposition policy by object type/classification and SHALL audit disposition actions.
Rationale	Implements data minimisation lifecycle.
Verification	Preview and execute test disposition with audit evidence.
Priority	MVP / P0
Traceability	F-ADM-003; PRI-02

SRS-FR-SEC-001 — OIDC and MFA

Requirement	The system SHALL authenticate via OIDC-compatible identity provider and enforce MFA according to deployment policy.
Rationale	Strong identity control.
Verification	Disable IdP user and verify access revoked; validate MFA path.
Priority	MVP / P0
Traceability	F-SEC-001

SRS-FR-SEC-002 — Role/object access

Requirement	The system SHALL enforce role, case membership, classification, and need-to-know at the API/service layer.
Rationale	Core confidentiality requirement.
Verification	Direct API access to unauthorised object must return denial regardless of UI.
Priority	MVP / P0
Traceability	F-SEC-002; SEC-01

SRS-FR-SEC-003 — Protected-source compartment

Requirement	The system SHALL store protected-source identity under separate authorization so analysts may use source-derived evidence without automatically seeing identity.
Rationale	Protects vulnerable sources.
Verification	Analyst can access derived evidence but not source identity.
Priority	MVP / P0
Traceability	F-SEC-003; SEC-02

SRS-FR-AUD-001 — Immutable audit trail

Requirement	The system SHALL record material create/update/delete/merge/unmerge/review/export/access-policy actions and ordinary users SHALL NOT modify audit records.
Rationale	Supports accountability.
Verification	Attempt audit modification as ordinary user and verify denial.
Priority	MVP / P0
Traceability	F-AUD-001

SRS-FR-OPS-001 — Backup and restore

Requirement	The system SHALL back up canonical DB, evidence, configuration, and required keys/metadata using documented recovery procedures and SHALL support periodic restore tests.
Rationale	Provides recoverability.
Verification	Perform restore drill into isolated environment and document result.
Priority	MVP / P0
Traceability	F-OPS-001

7. Data Requirements

SRS-DR-001 — Canonical object set

Requirement	The system SHALL implement at minimum Case, Source, EvidenceItem, EvidenceExtract, Fact, Entity, Relationship, Asset, Event, ValueFlow, Indicator, TypologyMatch, Hypothesis, IntelligenceGap, Assessment, IntelligenceProduct, Review, Dissemination, and AuditEvent as durable domain objects.
Rationale	Aligns SRS with canonical data model.
Verification	Schema review and CRUD contract tests for all required object types.
Priority	MVP
Traceability	Data Model v0.1

SRS-DR-002 — Stable identifiers

Requirement	Every canonical object SHALL have a stable, non-recycled identifier independent of display name.
Rationale	Supports audit and linking.
Verification	Rename object and verify identifier unchanged.
Priority	MVP
Traceability	Data Model v0.1

SRS-DR-003 — Version semantics

Requirement	Material assessments, charters, products, vocabularies, and other configured material objects SHALL preserve version history.
Rationale	Prevents historical overwrite.
Verification	Update object and retrieve prior version.

Priority	MVP
Traceability	AUD-01

SRS-DR-004 — Temporal semantics

Requirement	Objects with temporal meaning SHALL support valid-time fields and precision/unknown semantics rather than forcing exact dates.
Rationale	Supports imperfect open-source data.
Verification	Store year-only and approximate event date.
Priority	MVP
Traceability	Data Model v0.1

SRS-DR-005 — Provenance references

Requirement	Material facts, relationships, asset attributions, value-flow legs, indicators, and assessments SHALL reference supporting evidence and/or source objects.
Rationale	Core evidentiary traceability.
Verification	Run orphan-proposition integrity check.
Priority	MVP
Traceability	SRC-01; REL-01; VAL-01

SRS-DR-006 — Deletion semantics

Requirement	Deletion of canonical investigative objects SHALL be soft-delete/tombstone or equivalent where audit/history obligations require preservation; hard deletion SHALL be controlled by retention/disposition policy.
Rationale	Balances history with minimisation.
Verification	Delete test object and verify policy-compliant behaviour.
Priority	MVP
Traceability	PRI-02; AUD-01

8. External Interface Requirements

SRS-IF-001 — Web user interface

Requirement	The MVP SHALL provide a responsive web interface for supported desktop browsers; mobile support MAY be read-only or limited.
Rationale	Primary analyst workstation assumption.
Verification	Browser compatibility and viewport tests.

Priority	MVP
Traceability	PRD UX

SRS-IF-002 — REST/HTTP API

Requirement	The system SHALL expose versioned authenticated APIs for canonical objects and SHALL enforce identical authorization rules to the UI.
Rationale	Enables integration and testability.
Verification	API contract tests plus authorization parity tests.
Priority	MVP
Traceability	Technology Architecture

SRS-IF-003 — OIDC identity provider

Requirement	The system SHALL integrate with an OIDC-compatible provider for authentication and group/claim mapping.
Rationale	Avoids custom password storage where possible.
Verification	IdP login/logout/disable/claim tests.
Priority	MVP
Traceability	SEC-01

SRS-IF-004 — Object/evidence storage

Requirement	The system SHALL support encrypted object/file storage for evidence and derivatives with immutable-original semantics.
Rationale	Separates binary evidence from transactional records.
Verification	Upload/download/hash/permission tests.
Priority	MVP
Traceability	EVD-01

SRS-IF-005 — Export formats

Requirement	The system SHALL support PDF and machine-readable JSON/CSV where relevant for approved exports; exported analytical semantics SHALL preserve uncertainty/status/classification.
Rationale	Supports human and system recipients.
Verification	Round-trip test representative objects and inspect uncertainty fields.
Priority	MVP
Traceability	DIS-01

9. Security and Privacy Requirements

SRS-SEC-001 — Least privilege

Requirement	All authenticated access SHALL default deny and be granted by explicit role/object policy.
Rationale	Sensitive investigations require compartmentalisation.
Verification	Permission matrix tests including negative cases.
Priority	MVP
Traceability	SEC-01

SRS-SEC-002 — Encryption in transit

Requirement	All production HTTP/API traffic SHALL use TLS; plaintext administrative or application access SHALL be disabled except explicitly isolated development environments.
Rationale	Protects credentials and case data.
Verification	TLS configuration scan.
Priority	MVP
Traceability	Technology Architecture

SRS-SEC-003 — Encryption at rest

Requirement	Canonical databases, evidence stores, backups, and protected-source data SHALL use encryption at rest appropriate to deployment risk.
Rationale	Protects stolen media/snapshots.
Verification	Configuration and restore tests.
Priority	MVP
Traceability	SEC-02

SRS-SEC-004 — Secrets management

Requirement	Application secrets SHALL NOT be embedded in source code or container images and SHALL be stored/rotated through an approved secret-management mechanism.
Rationale	Reduces credential leakage.
Verification	Repository/image scan and rotation test.
Priority	MVP
Traceability	Technology Architecture

SRS-SEC-005 — Session control

Requirement	Sessions SHALL support idle timeout, absolute lifetime,
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	revocation, and reauthentication for configured high-risk actions.
Rationale	Reduces account takeover impact.
Verification	Session expiry/revocation test.
Priority	MVP
Traceability	SEC-01

SRS-SEC-006 — Protected source isolation

Requirement	Protected-source identity SHALL be logically separated from routine case content and SHALL use a narrower permission set.
Rationale	Protects human sources.
Verification	Cross-role authorization test.
Priority	MVP
Traceability	SEC-02

SRS-SEC-007 — Data minimisation

Requirement	The system SHALL support classification, purpose/scope linkage, retention state, and controlled export so sensitive data can be minimised throughout the lifecycle.
Rationale	Implements CSO privacy guardrails.
Verification	Retention/export/minimisation workflow test.
Priority	MVP
Traceability	PRI-01; PRI-02

SRS-SEC-008 — Security logging

Requirement	Security logs SHALL capture authentication, authorization failures, admin changes, and export events without unnecessarily recording evidence payloads or protected-source content.
Rationale	Balances detection and confidentiality.
Verification	Log inspection with synthetic sensitive fields.
Priority	MVP
Traceability	AUD-01

10. Audit and Provenance Requirements

SRS-AUD-001 — Audit event content

Requirement	Material audit events SHALL include actor, action, object
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	ID/type, timestamp, case/context when applicable, and before/after or change reference where appropriate.
Rationale	Allows reconstruction.
Verification	Inspect representative audit records.
Priority	MVP
Traceability	AUD-01

SRS-AUD-002 — Audit integrity

Requirement	Ordinary application users SHALL NOT alter or delete audit events. Administrative maintenance SHALL itself be logged.
Rationale	Prevents tampering.
Verification	Role test and admin-maintenance audit test.
Priority	MVP
Traceability	AUD-01

SRS-AUD-003 — Provenance traversal

Requirement	The system SHALL provide forward/backward traversal between Source, Evidence, derived artefacts, Facts, Indicators, Hypotheses, Assessments and Products.
Rationale	Makes reasoning reviewable.
Verification	Traverse complete pilot-case chain.
Priority	MVP
Traceability	ASM-003; EVD-02

SRS-AUD-004 — Merge history

Requirement	Entity merge/unmerge decisions SHALL record rationale, actor, timestamp, compared features, supporting evidence, and resulting mapping.
Rationale	Critical identity-control history.
Verification	Merge/unmerge audit test.
Priority	MVP
Traceability	ENT-01

11. Performance and Scalability

SRS-NFR-PERF-001 — Interactive latency

Requirement	For a reference MVP deployment, 95% of ordinary authenticated page/API reads SHOULD complete within 2
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	seconds excluding large file transfer, complex graph traversal, OCR, exports, and external connectors.
Rationale	Usability baseline.
Verification	Load test reference dataset.
Priority	MVP target
Traceability	PRD NFR

SRS-NFR-PERF-002 — Search latency

Requirement	Permission-filtered search SHOULD return first-page results within 3 seconds at reference MVP scale.
Rationale	Analyst productivity.
Verification	Search benchmark under concurrent users.
Priority	MVP target
Traceability	CAP-10

SRS-NFR-PERF-003 — Upload size

Requirement	The deployment SHALL define configurable maximum upload size and SHALL fail safely without partial canonical records when exceeded or interrupted.
Rationale	Evidence reliability.
Verification	Boundary upload tests.
Priority	MVP target
Traceability	F-DOC-001

SRS-NFR-PERF-004 — Reference scale

Requirement	MVP SHALL support at least 25 concurrent authenticated users, 100,000 canonical entities/relationships combined, and 100 GB evidence storage in a reference small-organisation deployment, subject to deployment profile tuning.
Rationale	Provides engineering sizing target.
Verification	Synthetic scale/load test.
Priority	MVP target
Traceability	Technology Architecture

12. Reliability, Backup and Disaster Recovery

SRS-NFR-REL-001 — Transactional consistency

Requirement	Canonical state changes that span related records SHALL
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	use transactional or compensating mechanisms to avoid orphan/inconsistent material records.
Rationale	Protects data integrity.
Verification	Fault-injection transaction tests.
Priority	MVP
Traceability	Data Model

SRS-NFR-REL-002 — Backup coverage

Requirement	Backups SHALL include canonical DB, evidence metadata and files, configuration required for restore, and protected-source data according to classification.
Rationale	Complete recovery.
Verification	Backup inventory review.
Priority	MVP
Traceability	OPS-001

SRS-NFR-REL-003 — Restore testing

Requirement	A restore procedure SHALL be executable into an isolated environment and SHOULD be tested at least quarterly for production deployments.
Rationale	Proves recoverability.
Verification	Documented restore drill.
Priority	MVP
Traceability	OPS-001

SRS-NFR-REL-004 — RPO/RTO profile

Requirement	Deployment documentation SHALL declare target RPO and RTO. MVP reference target SHOULD be RPO $\leq 24h$ and RTO $\leq 8h$ unless organisational risk requires stricter objectives.
Rationale	Makes resilience measurable.
Verification	DR plan review and timed restore.
Priority	MVP
Traceability	Technology Architecture

13. Search, Graph and Analytical Requirements

SRS-AN-001 — Canonical projection

Requirement	Search and graph stores SHALL be rebuildable projections from canonical objects and SHALL retain canonical object
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	identifiers.
Rationale	Avoids projection becoming truth source.
Verification	Delete/rebuild projection and compare object identity.
Priority	MVP/P1 where noted
Traceability	Technology Architecture

SRS-AN-002 — Uncertainty propagation

Requirement	Graph/search/analytics views SHALL display or expose material status/confidence/class fields used by the canonical records.
Rationale	Prevents visual certainty inflation.
Verification	Inspect disputed entity and reconstructed flow in all views.
Priority	MVP/P1 where noted
Traceability	Technology Architecture

SRS-AN-003 — Path semantics

Requirement	Any path/graph traversal result SHALL be presented as a path over existing relationships, not as proof of a direct relationship between endpoints.
Rationale	Prevents inference misuse.
Verification	Path-result wording/UI test.
Priority	MVP/P1 where noted
Traceability	TEC-02

14. AI and Automation Requirements

SRS-AI-001 — Derived status

Requirement	AI/OCR/NLP-generated outputs SHALL be marked as derived/candidate material with model/tool version and source context when material.
Rationale	Prevents automation from becoming evidence.
Verification	Generate candidate and inspect metadata.
Priority	P1 unless AI disabled
Traceability	TEC-01

SRS-AI-002 — Human verification

Requirement	AI output SHALL NOT become a material fact, canonical entity merge, adverse assessment, or dissemination approval without explicit human verification.
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Rationale	Preserves accountability.
Verification	Attempt automated promotion and verify blocked state.
Priority	P1 unless AI disabled
Traceability	TEC-01

SRS-AI-003 — Authorization boundary

Requirement	AI services SHALL receive only content the invoking user/service is authorised to access; retrieval for AI SHALL enforce the same object policies.
Rationale	Prevents cross-case leakage.
Verification	Prompt/AI retrieval authorization tests.
Priority	P1 unless AI disabled
Traceability	SEC-01

SRS-AI-004 — Optional feature flag

Requirement	AI assistance SHALL be disableable at deployment/environment level without breaking core investigative workflows.
Rationale	Supports high-risk/offline environments.
Verification	Disable AI and execute MVP pilot case.
Priority	P1 unless AI disabled
Traceability	F-ADM-004

15. UX and Accessibility Requirements

SRS-UX-001 — Evidence-first navigation

Requirement	Material analytical records SHALL expose a direct route to supporting evidence/source in no more than a small number of user actions.
Rationale	Review efficiency.
Verification	Usability test with analyst.
Priority	MVP
Traceability	PRD UX

SRS-UX-002 — Visible uncertainty

Requirement	Candidate/disputed entities, inferred relationships, and reconstructed/hypothetical flows SHALL be visually distinguishable from confirmed/direct records.
Rationale	Core analytical safety.

Verification	Visual regression/usability test.
Priority	MVP
Traceability	Technology Architecture

SRS-UX-003 — Destructive action confirmation

Requirement	Merge, unmerge, disposition, external export, and other high-impact actions SHALL require explicit confirmation and appropriate authorization.
Rationale	Reduces accidental harm.
Verification	Negative/confirmation tests.
Priority	MVP
Traceability	Control Guide

SRS-UX-004 — Accessibility baseline

Requirement	Primary workflows SHOULD meet WCAG 2.2 AA principles where feasible, including keyboard navigation, labels, contrast, and non-colour-only meaning.
Rationale	Inclusive use.
Verification	Automated plus manual accessibility checks.
Priority	MVP
Traceability	Product requirement

16. Administration and Configuration

SRS-ADM-001 — Policy configuration

Requirement	Admins SHALL configure classifications, role mappings, controlled vocabularies, retention rules, and feature flags without direct database editing.
Rationale	Safe administration.
Verification	Admin UI/API configuration tests.
Priority	MVP
Traceability	F-ADM-*

SRS-ADM-002 — Configuration audit

Requirement	Material policy/configuration changes SHALL be versioned or audit-logged with actor/time.
Rationale	Change accountability.
Verification	Change policy and inspect audit.

Priority	MVP
Traceability	AUD-01

SRS-ADM-003 — No silent retroactive rewrite

Requirement	Vocabulary/template changes SHALL NOT silently rewrite historical analytical meaning.
Rationale	Preserves historical context.
Verification	Rename vocabulary and inspect old record semantics.
Priority	MVP
Traceability	F-ADM-001

17. Interoperability and Integration

SRS-INT-001 — Connector provenance

Requirement	Any external connector SHALL retain provider/source, remote identifier, retrieval timestamp, mapping version, and licence/terms metadata where relevant.
Rationale	Makes enrichment attributable.
Verification	Connector fixture import test.
Priority	P1 except architectural rule
Traceability	SRC-01

SRS-INT-002 — No silent merge

Requirement	Imported external entity candidates SHALL NOT silently merge into canonical entities solely because an external identifier or fuzzy match is returned.
Rationale	Prevents external false positives.
Verification	Import ambiguous candidate and verify review required.
Priority	P1 except architectural rule
Traceability	ENT-01

SRS-INT-003 — FollowTheMoney mapping

Requirement	Where implemented, FollowTheMoney import/export SHALL preserve unmapped CS-AML fields and mapping-version metadata.
Rationale	Maintains interoperability without semantic loss.
Verification	Round-trip mapping test.
Priority	P1 except architectural rule

Traceability	F-INT-003
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SRS-INT-004 — Projection connectors

Requirement	Graph/search integrations SHALL be rebuildable and SHALL not own unique canonical analytical facts.
Rationale	Architectural consistency.
Verification	Projection rebuild test.
Priority	P1 except architectural rule
Traceability	Technology Architecture

18. Deployment and Operations

SRS-OPS-001 — Containerised/reference deployment

Requirement	A reference deployment SHOULD be reproducible using containerised or equivalent declarative deployment artefacts with environment-specific secrets externalised.
Rationale	Repeatable operations.
Verification	Fresh deployment from documentation.
Priority	MVP
Traceability	Technology Architecture

SRS-OPS-002 — Health endpoints

Requirement	Application services SHALL provide health/readiness indicators that do not reveal case-sensitive content.
Rationale	Operational monitoring.
Verification	Health endpoint test.
Priority	MVP
Traceability	OPS

SRS-OPS-003 — Migration safety

Requirement	Schema migrations SHALL be versioned, reversible where practical, and tested against representative backup data before production application.
Rationale	Protects canonical records.
Verification	Migration dry-run/rollback test.
Priority	MVP
Traceability	Data Model

SRS-OPS-004 — Observability minimisation

Requirement	Metrics/logs/traces SHALL avoid storing evidence bodies, protected-source identity, or other case-sensitive payloads unless explicitly required and protected.
Rationale	Prevents telemetry leakage.
Verification	Telemetry inspection.
Priority	MVP
Traceability	SEC-02

19. Verification and Acceptance

Each mandatory requirement SHALL be verifiable by one or more of: automated unit/integration/API test, security test, data-integrity test, UI acceptance test, operational drill, or structured manual review. A requirement is not complete merely because code exists.

Code	Verification method
V1	Automated functional/API test
V2	Authorization/security negative test
V3	Data integrity/schema/provenance test
V4	UI/usability acceptance test
V5	Operational backup/restore/deployment drill
V6	Structured reviewer sign-off

20. Traceability Matrix

SRS family	Upstream product	Framework/control	Canonical data	Primary test
CASE	F-CASE-*	GOV-01/CAS-01/CAS-02	Case, Review, AuditEvent	V1/V6
EVD	F-EVD-*/F-DOC-001	SRC-01/SRC-02/EVD-01/ EVD-02	Source, EvidenceItem, EvidenceExtract	V1/V3
ENT/REL/AST	F-ENT-*/F-REL-*/F- AST-001	ENT-01/REL-01/AST-01	Entity, Relationship, Asset	V1/V3/V4
TIM/VAL	F-TIM-*/F-VAL-*	REL-01/VAL-01	Event, ValueFlow	V1/V3/V4
TYP/HYP/ASM	F-TYP-*/F-HYP-*/F-ASM-*	TYP-01/HYP-01/HYP-02/ ASM-01/GAP-01	Indicator, TypologyMatch, Hypothesis, Assessment	V1/V4/V6
PRD/REV/DIS	F-PRD-*/F-REV-001/F-DIS- *	QUA-01/DIS-01/PRI-01	IntelligenceProduct, Review, Dissemination	V1/V2/V6
SEC/AUD/OPS	F-SEC-*/F-AUD-001/F- OPS-001	SEC-01/SEC-02/AUD-01/ PRI-02	AccessPolicy, AuditEvent, BackupRecord	V2/V5

21. MVP Release Baseline

Release gate

MVP 0.1 SHALL NOT be declared production-ready until the end-to-end pilot scenario passes with provenance, authorization, audit, review, and export controls intact.

- A case can be opened only with accountable owner and valid charter.
- Evidence originals and derivatives are separable and integrity-protected.
- At least one entity merge/unmerge is demonstrated without history loss.
- Ownership/control/asset and mixed-class value flows are represented without semantic collapse.
- At least two competing hypotheses are tested using supporting and contradicting evidence.
- An assessment is confidence-rated and backward traceable to sources.
- Independent peer review is completed.
- External export is blocked until dissemination approval.
- Audit trail captures all material actions.
- Backup restore drill succeeds in an isolated environment.

Annex A — Requirement ID Taxonomy

- SRS-FR-* Functional requirements
- SRS-DR-* Data requirements
- SRS-IF-* External interfaces
- SRS-SEC-* Security/privacy
- SRS-AUD-* Audit/provenance
- SRS-NFR-PERF-* Performance
- SRS-NFR-REL-* Reliability
- SRS-AN-* Search/graph/analytics
- SRS-AI-* AI/automation
- SRS-UX-* UX/accessibility
- SRS-ADM-* Administration
- SRS-INT-* Integration
- SRS-OPS-* Deployment/operations

Annex B — State Models

Case: DRAFT → TRIAGE → ACTIVE → REVIEW → APPROVED/CLOSED → MONITORING → REOPENED

Entity resolution: CANDIDATE → PROBABLE → CONFIRMED | DISPUTED | UNRESOLVED

Intelligence product: DRAFT → IN_REVIEW → CHANGES_REQUESTED → APPROVED → DISSEMINATED → SUPERSEDED/CORRECTED

Annex C — API Resource Baseline

- /cases
- /sources
- /evidence
- /evidence-extracts
- /entities
- /relationships
- /assets
- /events
- /value-flows
- /indicators
- /typologies
- /hypotheses
- /gaps
- /assessments
- /products
- /reviews
- /disseminations
- /audit-events
- /search

API rule

Resource naming is informative; exact REST/GraphQL design may vary. Authorization semantics, stable identifiers, provenance, and uncertainty preservation are normative.

Annex D — Test Scenario Baseline

1. Create a high-sensitivity case and charter.
2. Register two public sources and upload three evidence files; hash originals.
3. Create evidence extracts and one OCR derivative.
4. Create two similar company records, evaluate candidate match, merge, then unmerge one test cycle.
5. Create Person, Company, Property and Contract entities plus ownership/control relationships.
6. Add timeline events for incorporation, contract award, property acquisition and ownership change.
7. Create a four-leg value flow containing DIRECT, DOCUMENTED, RECONSTRUCTED and HYPOTHETICAL legs.
8. Map two indicators and one counter-indicator to a typology.

9. Create two competing hypotheses and record supporting/contradicting evidence.
10. Create assessment with Moderate confidence and two intelligence gaps.
11. Generate intelligence product, submit to independent reviewer, address change request, approve.
12. Attempt export before approval (must fail), then approve dissemination and export.
13. Verify sharing log and audit history.
14. Restore the case from backup into isolated environment and verify object/evidence hashes.